

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1366

Omro 138 kV
July 18, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1366. The Interconnection Customer proposes a 500.3 MW solar photovoltaic facility interconnecting at the Omro 138 kV substation in the Pennsylvania area. The request was received on December 25, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 419 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$89,364,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Omro – Clrk Rd 138 KV	William F Matson Generating Station 138/1 kV Transformer	200.2	225	100.0
Cox Battery Energy Storage – William F Matson Generating Sta 138 KV	Clark Rd – Omro 138 KV	176.7	255	100.0
Aseboro – Clark Rd 138 KV	William F Matson Generating Station 138/1kV Auto	152.2	206	65.3
Axford – Empire 138 KV	Southbase – Milwaukee 138 KV	140.9	273	6.9
Asheboro – Empre 138 KV	Southbase – Milwaukee 138 KV	137.4	283	6.9
Rewey 345 kV/138 kV Transformer	Orchard Rd – Ft Humbug 500 KV	136.0	446	11.4
William F Matson Generating Station Substation 138-1 kV Transformer Bank	Omro – Clark Rd Substation 138 KV line	125.3	360	100.0
Brent – Axford 138 KV	Milwaukee Substation – Southbase 138 KV	122.5	312	6.9
Omro – Searsburg 138 KV Line	Omro – Clark Rd 138 KV Line	117.1	227	54.4
Cox Battery Energy Storage – Locusst 138 KV	Omro – Searsburg 138 KV	113.1	218	60.7
Hamilton Rd Substation – Edes Tap 2 138 KV	Edes Tap 2 – Lutesville 345 KV	110.0	271	5.7
William F Matson Generating Sta Substation 345/1kV Autotransformer	Omro – Clark Rd Substation 138 KV	110.0	410	100.0
Hamilton Rd – Wien 138 KV line	Edes Tap 2 – Lutesville 345 KV	109.0	285	5.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Warrior Ridge Hydro – Wien 138 KV line	Lutesville – Edes Tap 2 Substation 345 KV line	108.8	288	5.7
Cox Battery Energy Storage – Searsburg Substation 138 KV	Omro – Clark Rd 138 KV	103.9	220	48.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omro – Clark Rd 138 KV line	Rebuild line with higher-capacity conductor	\$14,604,000	100% (\$14,604,000)
Cox Battery Energy Storage – William F Matson Generating Sta 138 KV L	Rebuild line with higher-capacity conductor	\$444,000	100% (\$444,000)
Aseboro – Clark Rd 138 KV	Rebuild line with higher-capacity conductor	\$18,822,000	94% (\$17,710,000)
Empire – Axford 138 KV	Rebuild line with higher-capacity conductor	\$2,748,000	34% (\$942,000)
Asheboro – Epire 138 KV line	Rebuild line with higher-capacity conductor	\$17,221,000	34% (\$5,904,000)
Rewey 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$31,786,000	40% (\$12,872,000)
William F Matson Generating Station 138/1kV Autotransformer	Replace transformer with higher-rated unit	\$5,754,000	100% (\$5,754,000)
Brent – Axford 138 KV	Rebuild line with higher-capacity conductor	\$4,357,000	34% (\$1,494,000)
Omro – Searsburg 138 KV Line	Rebuild line with higher-capacity conductor	\$1,615,000	100% (\$1,615,000)
Locusst – Cox Battery Energy Storage 138 KV	Rebuild line with higher-capacity conductor	\$4,086,000	100% (\$4,086,000)
Hamilton Rd Substation – Edes Tap 2 138 KV	Rebuild line with higher-capacity conductor	\$1,522,000	51% (\$779,000)
William F Matson Generating Sta Substation 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$18,065,000	100% (\$18,065,000)
Hamilton Rd – Wien 138 KV line	Reconductor line	\$2,316,000	51% (\$1,185,000)
Warrior Ridge Hydro – Wien Substation 138 KV	Reconductor line	\$3,306,000	51% (\$1,691,000)
Cox Battery Energy Storage – Searsburg Substation 138 KV	Reconductor line	\$2,219,000	100% (\$2,219,000)
	Total allocated to project		\$89,364,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.